MSBA7029: Storytelling with Data

Spring 2027 (Revised Syllabus — Draft)

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Background

It is common in business to have to explain complex analysis to audiences of varying technical levels. Storytelling with data is thus an essential skill in business, as is directing attention to what matters, and motivating the proper action. This course treats data storytelling as a discipline grounded in three pillars: the empirical science of graphical perception, formal frameworks for constructing visuals and narratives, and professional craft in explanatory communication.

Course Design Rationale

The course assumes familiarity with at least one tool (R, Python, or Excel) for producing standard visualizations for creating common visualizations such as scatterplots, bar charts, line plots, and heatmaps. Generative AI tools like ChatGPT make this prerequisite much easier to meet. The course is therefore agnostic to a specific tool in principle, though some Python familiarity is advantageous for exploring the AI tools.

As generative AI tools continue to evolve, they are becoming increasingly embedded in exploratory data analysis. Still, a solid foundation in best practices remains crucial to understand and properly modify the outputs of these AI. The focus of the course is therefore to understand the core principles of data visualization and exploratory data analysis in order to use these tools most effectively.

To maintain a steady study pace, each class will include a quiz. Students will have in-class practice and after-class ungraded assignments. The final group project will tie together all concepts covered in the course.

Course Objectives

1. **Ground design choices in evidence.** Explain and apply results from graphical perception research when choosing encodings and chart types.
2. **Construct visuals from first principles.** Use the grammar of graphics to derive, rather than look up, the appropriate visual for a given data structure and message.
3. **Communicate uncertainty honestly.** Select and defend appropriate representations of uncertainty for forecasts, experiments, and model outputs.
4. **Build decision-driving narratives.** Structure findings using the Big Idea, SCQA, and narrative visualization genres, tailored to specific business audiences.
5. **Detect and avoid deception.** Recognize distortion techniques, reason about visualization ethics, and use generative AI responsibly with appropriate provenance and disclosure.
6. **Communicate model results.** Present the performance and limitations of predictive models

to non-technical decision-makers.

Course Materials

There is no textbook to purchase. Some suggested references are listed below:

- Kieran Healy, *Data Visualization: A Practical Introduction*
- Claus Wilke, *Fundamentals of Data Visualization*
- Cole Nussbaumer Knaflic, *Storytelling with Data*

Course Schedule

Session	Topic	Content
1	**Context, Audience, and the Big Idea**	Explanatory vs. exploratory analysis; audience analysis; the Big Idea; storyboarding before software. Minto's Pyramid Principle and SCQA (Situation–Complication–Question–Answer)
2	**Graphical Perception: The Evidence Base**	Why some encodings work: Cleveland & McGill's ranking of elementary perceptual tasks; the Heer & Bostock crowdsourced replication; preattentive attributes; position vs. length vs. angle vs. area.
3	**A Grammar of Graphics; Choosing Visuals**	Chart choice: data, aesthetics, geometries, scales, facets. Knaflic's endorsed chart vocabulary. Ehrenberg's rules, ordering by magnitude, when a table beats a chart.
4	**Clutter, Attention, and Design**	Gestalt principles for decluttering; signal-to-noise; focusing attention with size, position, and colour; affordance and visual hierarchy.
5	**Color and Accessibility**	Perceptually uniform colormaps; sequential vs. diverging vs. qualitative palettes; colourblind-safe design; alt text, contrast, and WCAG basics for dashboards.
6	**Visualizing Uncertainty**	Error bars, confidence bands, gradient and fan charts; quantile dotplots; hypothetical outcome plots; why audiences misread the "cone of uncertainty"; communicating forecast and A/B test uncertainty.
7	**Deception, Distortion, and Data Ethics**	Truncated and dual axes; area-for-length errors; cherry-picked baselines and windows; Simpson's paradox rendered visually; maps, choropleths, and the MAUP; experimental evidence on deceptive visualization.
8	**Narrative Visualization and Story Structure**	Segel & Heer's genre taxonomy; martini glass, interactive slideshow, and drill-down structures; narrative arc, tension, and resolution; horizontal and vertical logic in a deck; reverse storyboarding as QA.

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Session	Topic	Content
9	**Generative AI for Explanatory Analysis**	LLMs as chart critics. Where multimodal models fail at reading charts; automated insight generation and the false-discovery problem; provenance, disclosure, and responsible use of genAI in client-facing work.
10	**Communicating Model Results**	Presenting classifier and model performance to non-technical decision-makers. Course integration and group project workshop.

Assessment

Component	Points	% of Overall Grade
Class Participation	20	20%
Individual Assignments	25	25%
Group Project	30	30%
Class Quizzes	25	25%
Total	**100**	**100%**

Class Participation (20%)

Based on contribution to lectures, after-class discussion, and above all the weekly critique studio. Each student presents at least one found chart during the term and contributes to redesign discussions. Short response memos for guest speaker sessions (one paragraph each) also count. Grading follows a rubric: presence alone earns at most half credit; the remainder reflects the quality and specificity of critique.

Individual Assignments (25%)

Each student submits three before/after redesigns across the term: an original chart found in the wild (business press, corporate reports, or the student's own workplace), a redesigned version, and a rationale of roughly 300 words that must cite specific principles from the course (perceptual evidence, grammar-of-graphics reasoning, uncertainty or ethics considerations). Submissions are staggered (approximately Sessions 4, 7, and 9) so feedback from each informs the next.

Group Project (30%)

The full project specification, including the dataset options, deliverables, rubric, and presentation date, is released in Session 1. Groups of four to five, formed voluntarily by Session 3. Deliverables: an executive presentation deck (presented live in the final week), a written memo in pyramid structure (maximum three pages), and a one-page appendix disclosing any generative AI use and verifying all figures against source data. Each group member will receive the same score.

Class Quizzes (25%)

A closed-book quiz in each of Sessions 2–10, plus a survey quiz in Session 1 (ten quizzes total; the lowest score is dropped). Quiz questions are drawn primarily from the empirical and framework material—perceptual rankings, grammar-of-graphics concepts, uncertainty representations, deception mechanisms—where answers are defensible from the readings, rather than from matters of aesthetic judgment. No make-ups except under irresistible circumstances (e.g., illness, job interviews), per HKU policy.

Academic Integrity and Generative AI Policy

Plagiarism is penalized according to HKU rules and guidelines. Generative AI tools may be used for brainstorming, drafting, and implementation in all assessed work except in-class quizzes, subject to two requirements: every submission must include a brief disclosure of how genAI was used, and students remain fully responsible for the accuracy of every figure, number, and claim submitted. Presenting unverified AI-generated analysis is treated as a professional failure and graded accordingly; the ethics of this boundary is itself course content (Sessions 7 and 9).